6.3 Implementing BDD more effectively with Thucydides

Thucydides (http://thucydides.info) is an open source library that adds better-integrated and more-comprehensive reporting capabilities to conventional BDD tools such as JBehave and Cucumber.

You've already briefly seen Thucydides in action in chapter 2, where it helped generate the test reports. When you use it with a fully supported BDD tool, it helps you break down scenario implementations into smaller reusable steps and report on these steps in the living documentation. At the time of writing, Thucydides works well with JVM-based testing libraries such as JBehave and JUnit, but you can also import test results generated by other BDD tools such as Cucumber-JVM,[4] SpecFlow, Cucumber-JS, or Behave.

For the fully integrated tools, Thucydides also makes writing automated acceptance tests with WebDriver, a popular open source browser automation library, easier and more productive.

But the specialty of Thucydides is taking the test results produced by BDD tools like JBehave and turning them into rich, well-integrated living documentation, such as by relating the test outcomes to the requirements they test. This gives stakeholders a clear picture of not only what tests have been executed, but what requirements have been tested, and how well they've been tested.

Even if you're not using one of the fully supported BDD tools, you can still benefit from many of the reporting and requirements integration features, and use Thucydides as a generator of living documentation. We'll look at Thucydides in much more detail when we discuss Living Documentation in chapter 11.

In the remainder of this chapter, we'll discuss specific BDD tools so you can get started using a BDD tool in your environment and trying out these techniques for yourself. The first tool we'll look at is JBehave.

6.4 Automating scenarios in Java with JBehave

JBehave (http://jbehave.org) is a popular Java-based BDD framework that was originally written by Dan North. In JBehave, you write step definition methods in Java or in other JVM languages such as Groovy or Scala.

We'll look at how to use JBehave in conjunction with Thucydides, as Thucydides supports a number of conventions that make setting up and configuring a JBehave project simpler.

6.4.1 Installing and setting up JBehave

The easiest way to build and run a JBehave/Thucydides test suite is to use Maven. Maven (http://maven.apache.org) is a widely used build tool in the Java world. Maven provides a number of features that simplify and standardize the build project for Java-based projects, including a standardized directory structure and build lifecycle, and powerful dependency management capabilities.

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[4] At the time of writing, full Cucumber integration was being actively developed.